Physical Layer : Data and Signal

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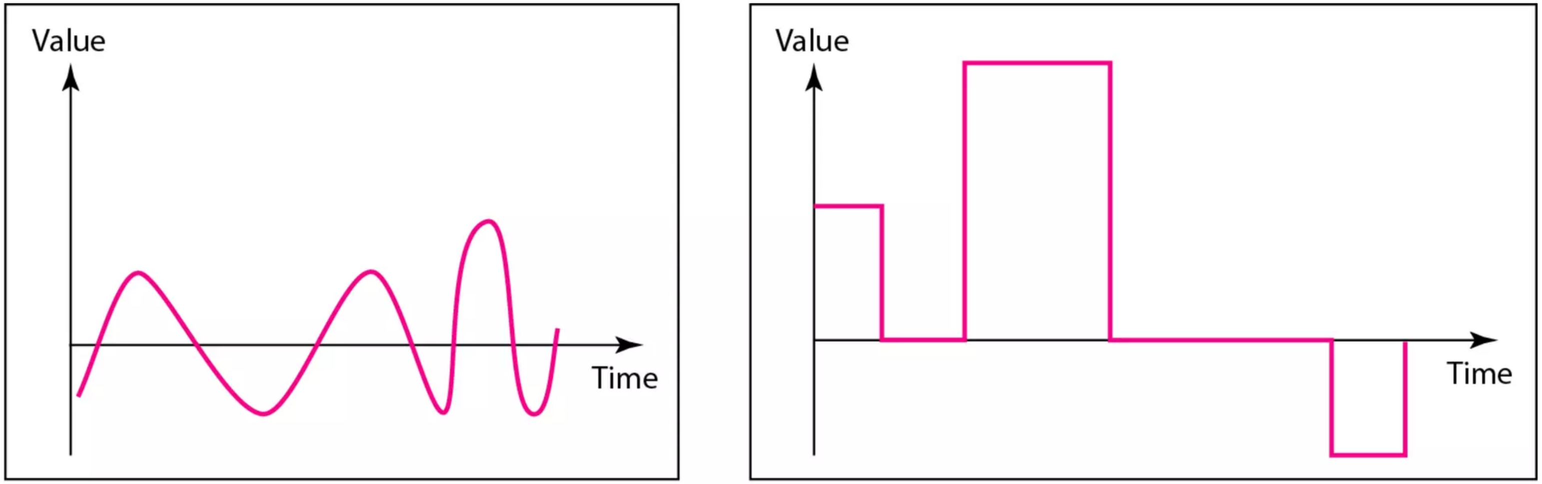
Signal

- The physical layer moves data in the form of electro-magnetic signals across a transmission medium
- Data can be transmitted on wires by varying some physical property such as **voltage** or **current**
- Data can be analog or digital
  - Analog data refers to information that is continuous
  - Digital data refers to information that has discrete states

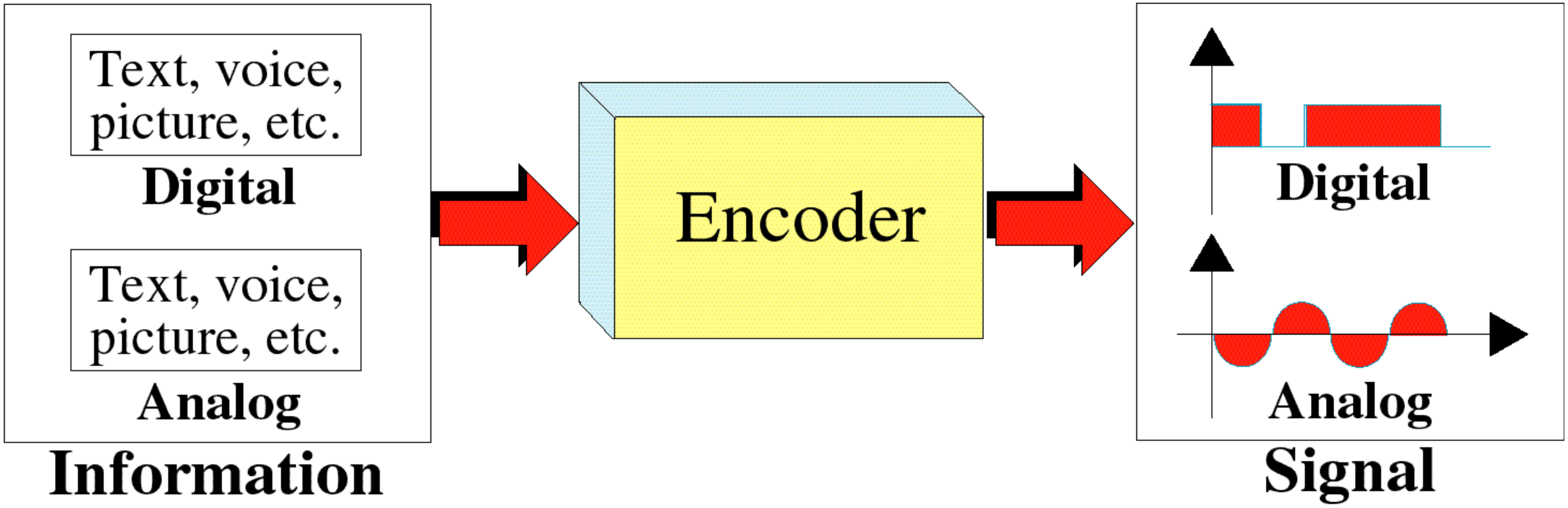
Analog and Digital Signal

- Signal also has either analog or digital in form
  - An analog signal has infinitely many levels of intensity over a period of time
  - A digital signal, on the other hand, can have only a limited number of defined values



a. Analog signal b. Digital signal

Transformation of Information to Signal



Periodic vs. non-periodic Signal

- Both analog and digital signals can be either periodic or nonperiodic (aperiodic)
- A periodic signal completes a pattern (period) within a measurable time frame and repeats that pattern over subsequent identical periods
- The completion of one full pattern is called a cycle
- A nonperiodic signal changes without exhibiting a pattern or cycle that repeats over time.
- In data communications, we commonly use periodic analog signals

Periodic analog signals

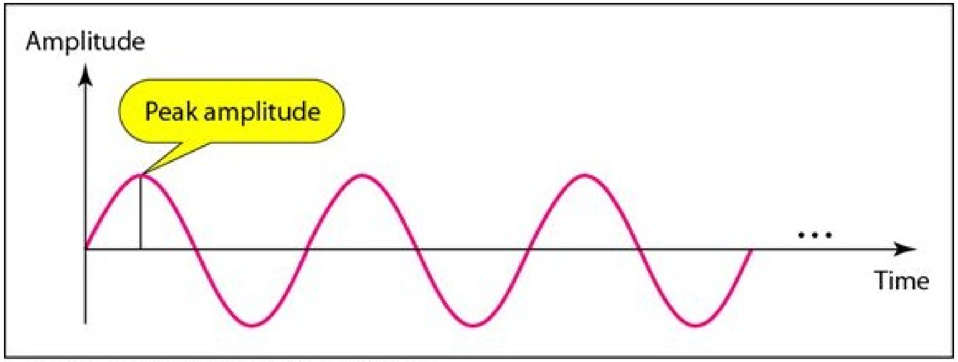
- Can be simple or composite
- The sine wave – simplest form of periodic analog signal
- Sine wave has general form :

$$g(t) = A.\sin(2\Pi ft+\Phi)$$

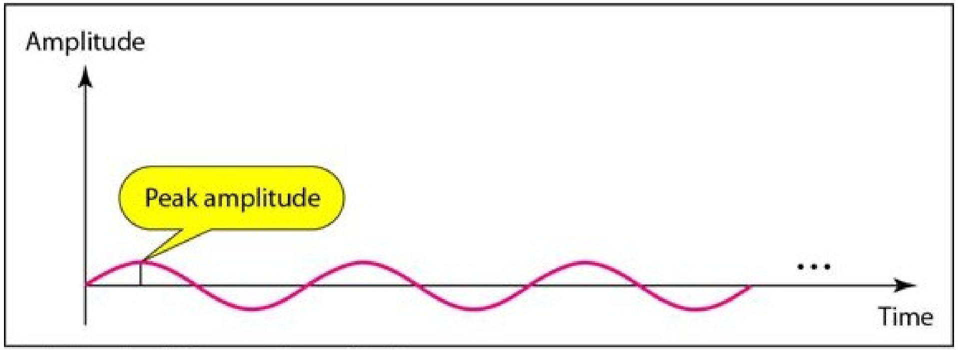
  - Where, A is the peak amplitude, f the frequency, t is the time, and Φ is the phase

Peak amplitude

- It is the absolute value of its highest intensity, proportional to the energy it carries.
- For electric signals, peak amplitude is normally measured in volts



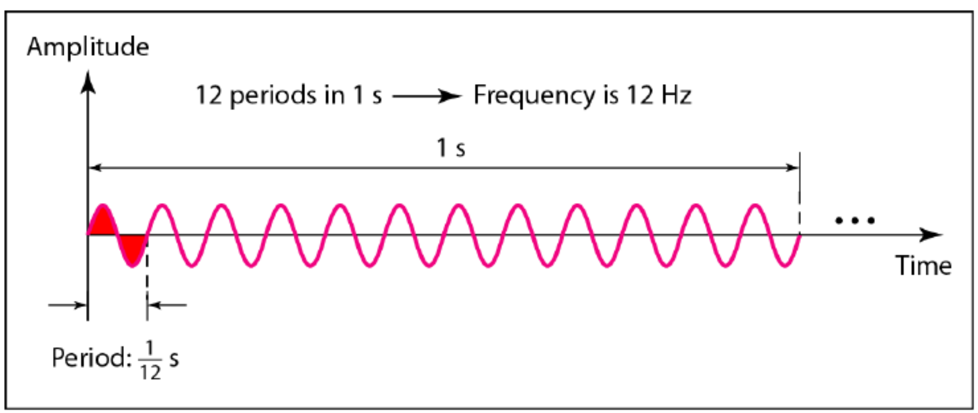
a. A signal with high peak amplitude

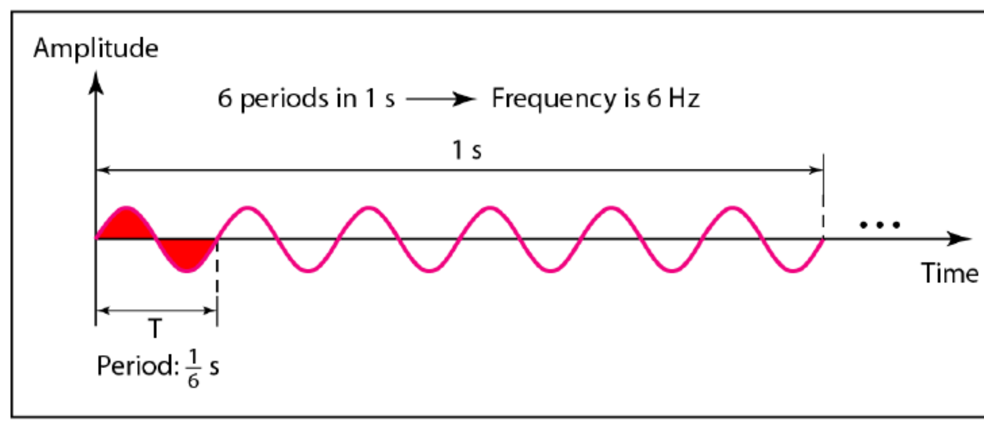


b. A signal with low peak amplitude

Period and Frequency

- Period (T) refers to the amount of time, in seconds, a signal needs to complete 1 cycle.
- Frequency (f) refers to the number of periods in one second
  - f=1/T and T=1/f
  - Frequency is formally expressed in Hertz (Hz) (cycles per second)
- Frequency - the rate of change with respect to time.
  - If a signal does not change at all, f = 0
  - If a signal changes instantaneously, f = ∞.





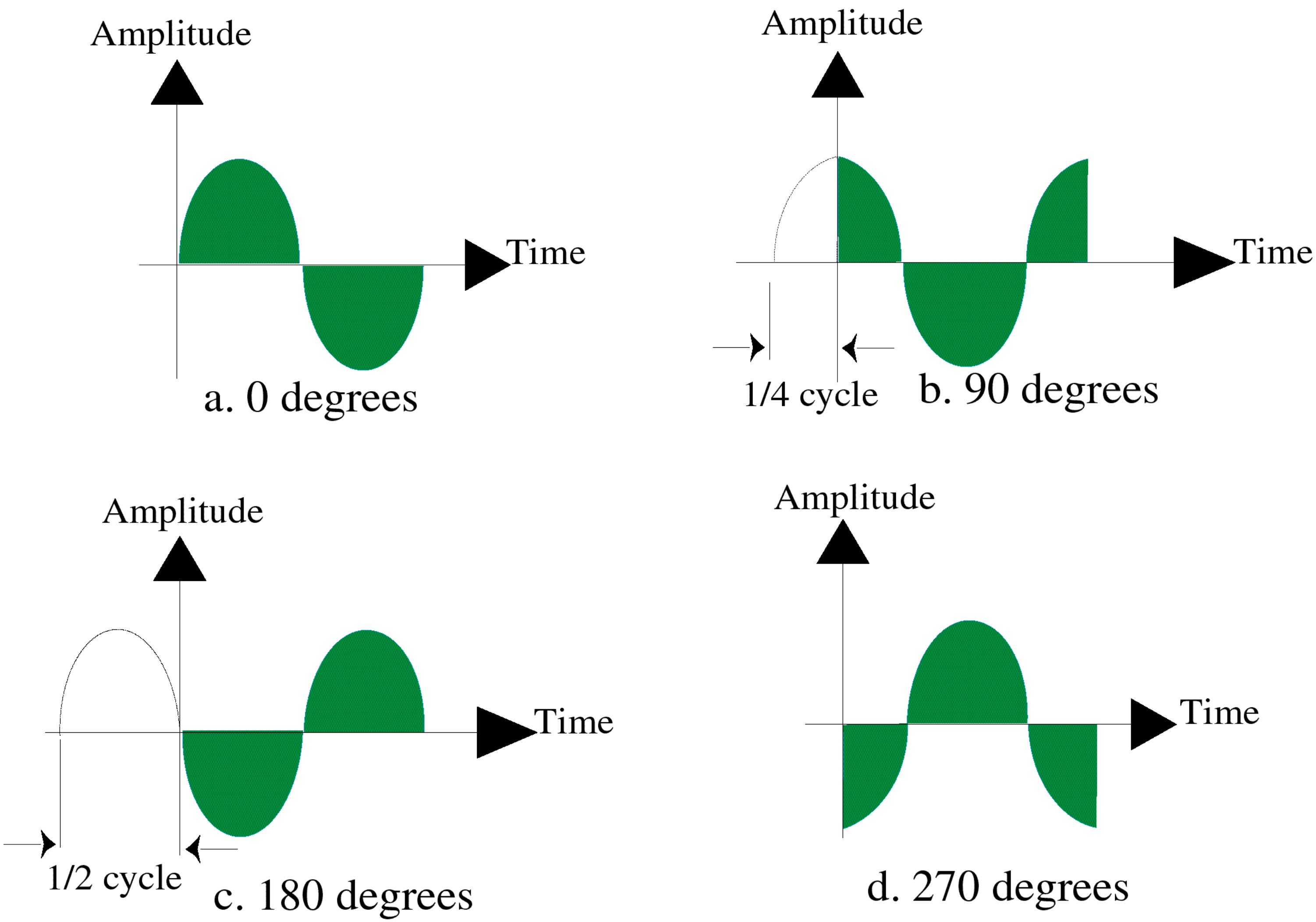
Period and Frequency

- The power we use at home has a frequency of 60 Hz. What is the period?
- The period of a signal is 100 ms. What is its frequency in kilohertz?

Phase

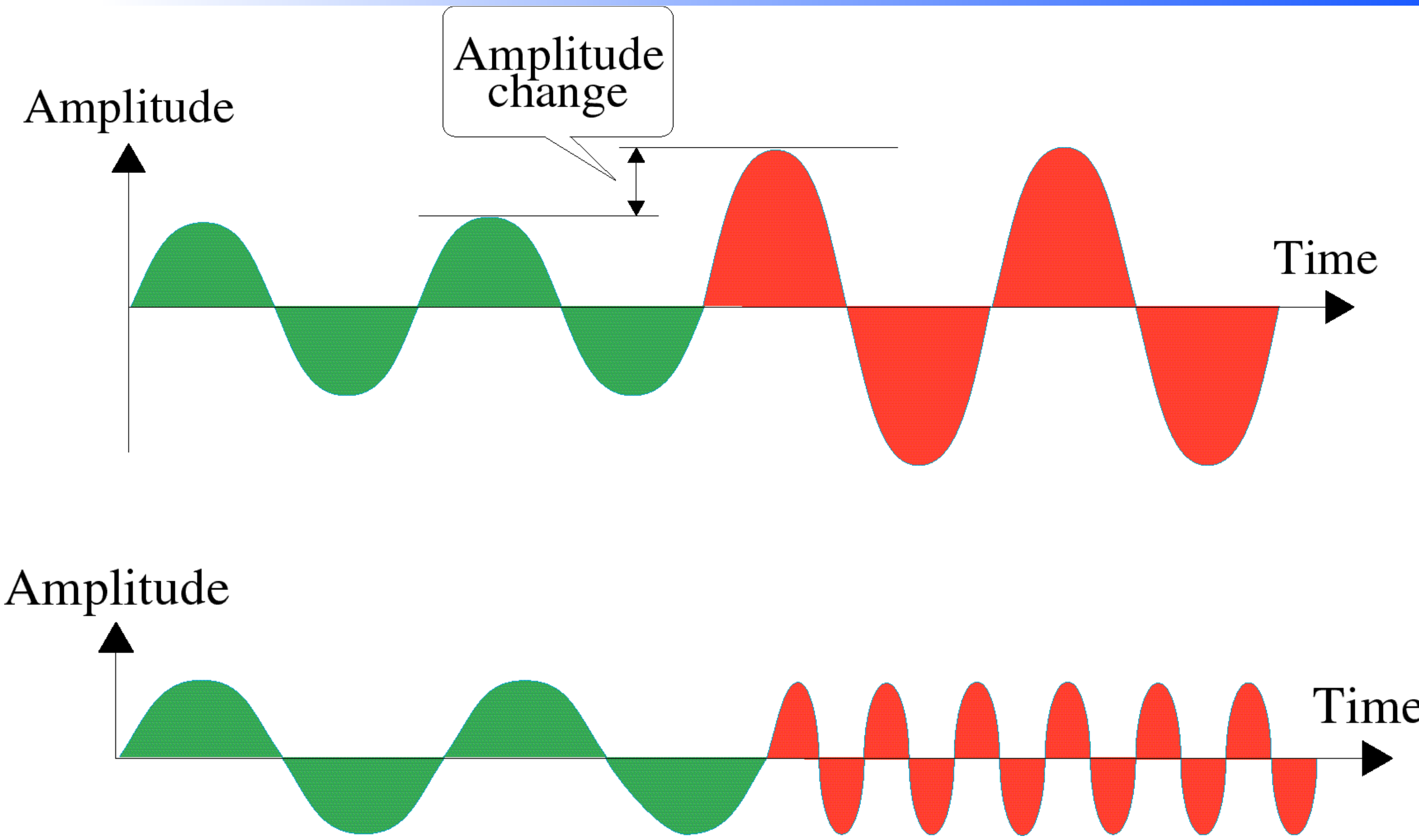
- The term phase describes the position of the waveform relative to time 0
- Phase describes the amount of shift
- Indicates status of the first cycle
- Phase is measured in degrees or radians

Phase

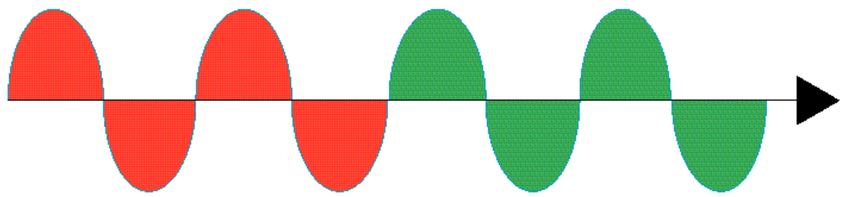


Amplitude
Time
a. 0 degrees
Amplitude
Time
1/4 cycle
b. 90 degrees
Amplitude
Time
1/2 cycle
c. 180 degrees
Amplitude
Time
d. 270 degrees

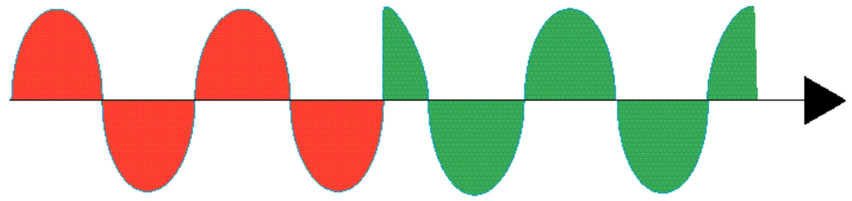
Change in Amplitude and Frequency



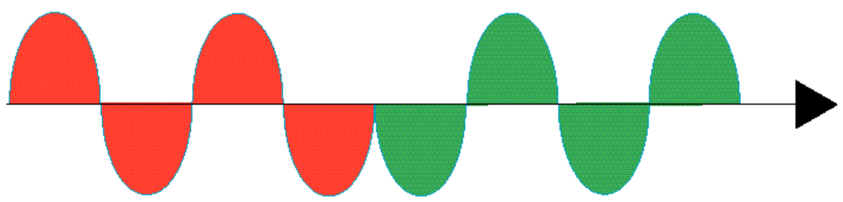
Change in phase



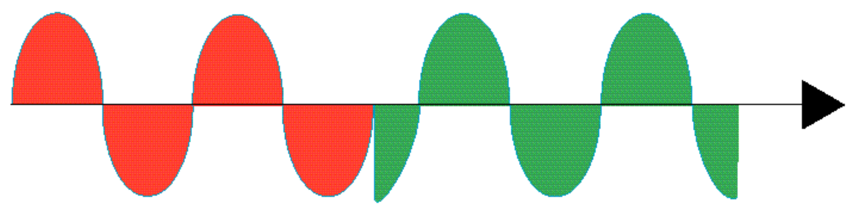
a. No phase change



b. 90 degree phase change



c. 180 degree phase change



d. 270 degree phase change

Wavelength

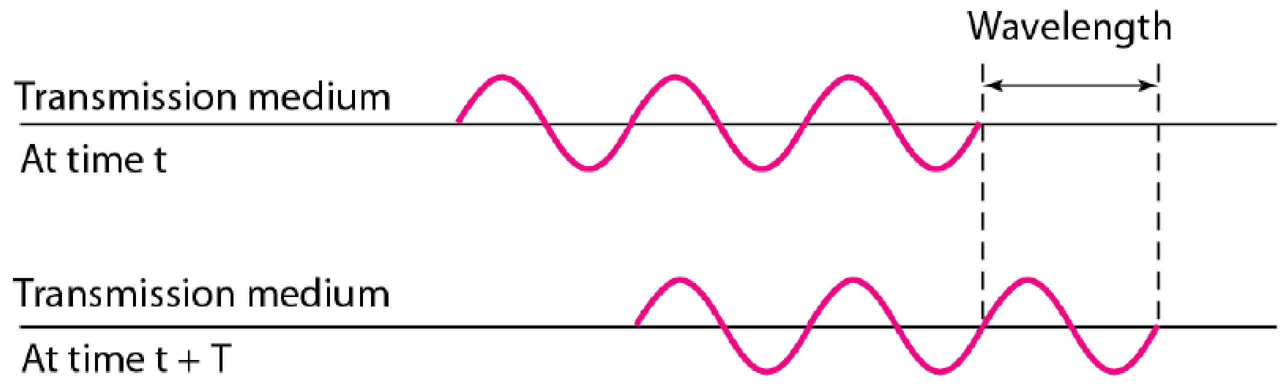
- It binds the period or the frequency of a simple sine wave to the propagation speed of the medium
- The wavelength depends on both the frequency and the medium
- In data communications, wavelength is used to describe the transmission of light in an optical fiber

Wavelength(2)

- The wavelength **is the distance a simple signal can travel in one period**

Wavelength = propagation speed X period = propagation speed / frequency

- The propagation speed of electromagnetic signals depends on the medium and on the frequency of the signal
- In a vacuum, light is propagated with a speed of **3×10^8 m/s**
- The wavelength is normally measured in micrometers (microns) instead of meters

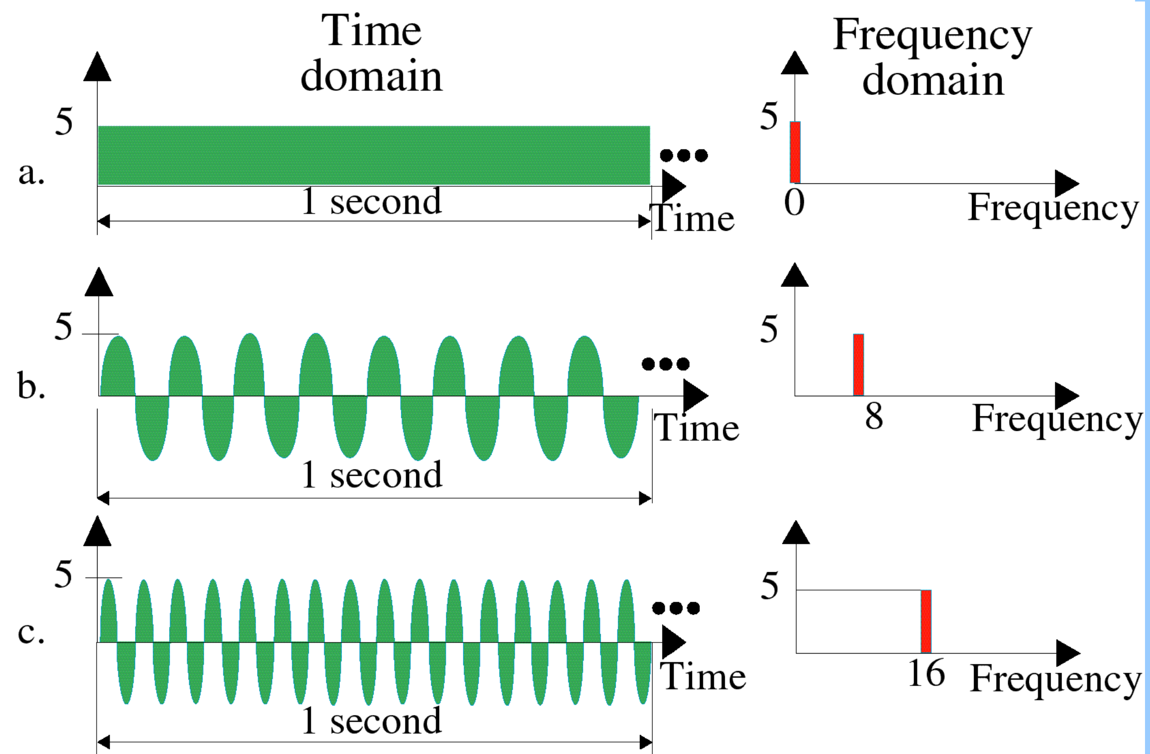


Wavelength

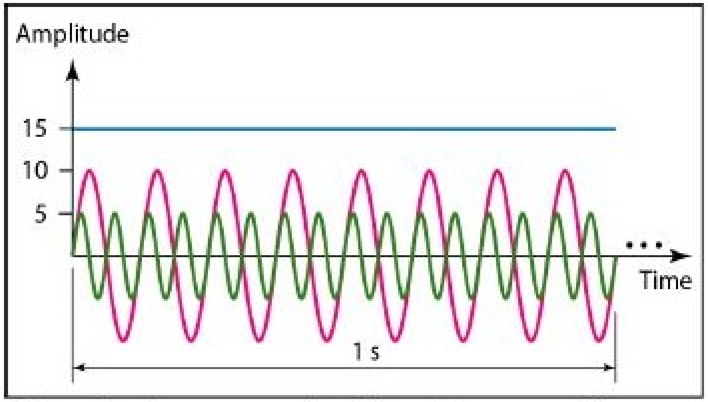
- What is the wavelength of red light ($f=4.28 \times 10^{14}$ Hz)?

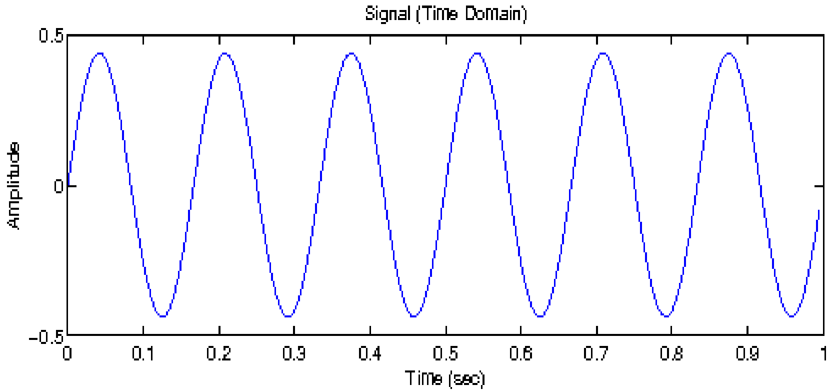
Time and Frequency domain

- The time-domain plot shows changes in signal amplitude with respect to time
- A frequency-domain plot is concerned with only the peak value and the frequency
- The advantage of the frequency domain is that we can immediately see the values of the frequency and peak amplitude



Draw Frequency Domain





Composite Signal

- Simple sine wave makes no sense as it can't carry any information
- We need to send a composite signal to communicate data. A
- According to **Scientist Fourier**, periodic function, g(t) with period T, can be constructed as the sum of a (possibly infinite) number of sines and cosines (with f - frequency, a_n and b_n are the sine and cosine amplitudes of the nth harmonics (terms), and a constant c)

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

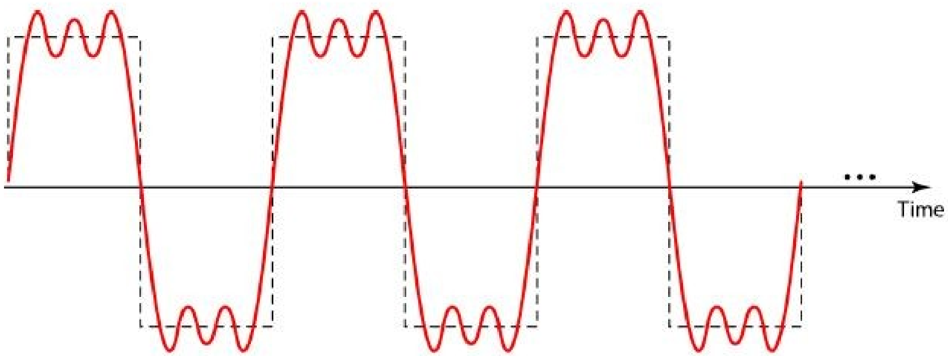
Composite Signal(2)

- Composite signal is made of many simple sine waves with different frequencies, amplitudes, and phases

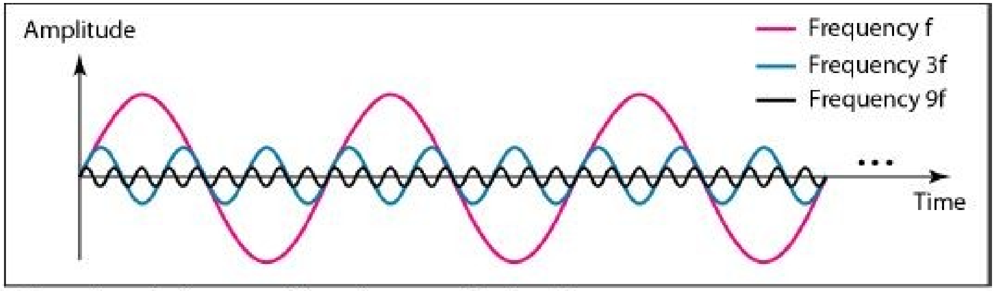
$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \sin(2\pi n f_0 t + \phi_n)$$

- A periodic composite signal can be decomposed into a series of simple sine waves with discrete frequencies(integer values 1, 2, 3, …)
- A nonperiodic composite signal can be decomposed into a combination of an infinite number of simple sine waves with continuous frequencies(real values)

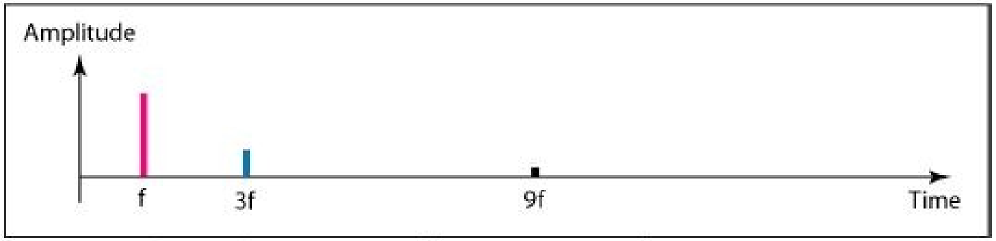
Example of Decomposition of composite signal



Periodic



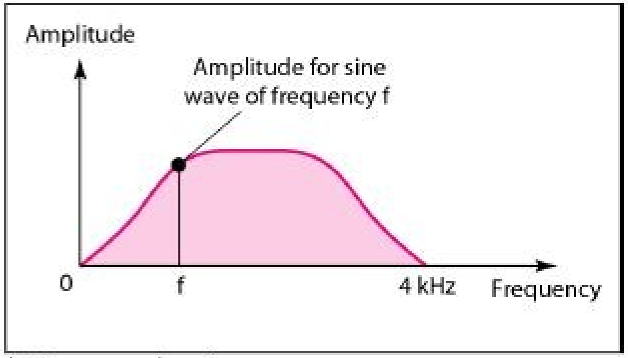
a. Time-domain decomposition of a composite signal



b. Frequency-domain decomposition of the composite signal

Aperiodic

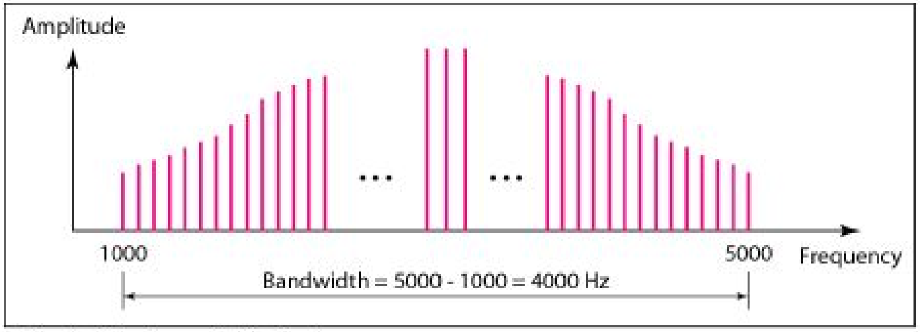
a. Time domain



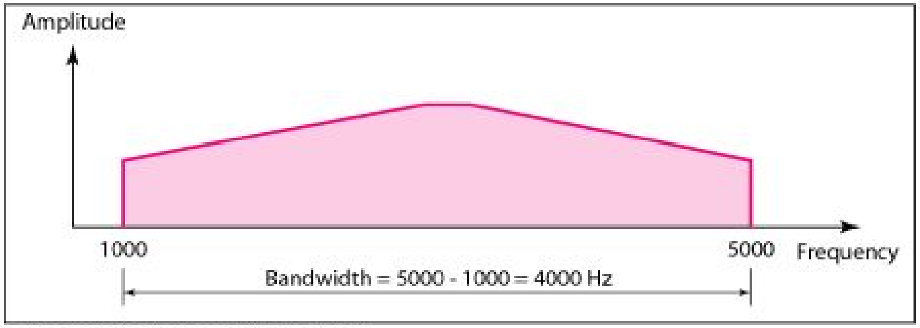
b. Frequency domain

Bandwidth

- The range of frequencies contained in a composite signal is its bandwidth.
- $B = f_h - f_l$
  - if a composite signal contains frequencies between 1000 and 5000 Hz, its bandwidth is 5000 - 1000, or 4000 Hz.



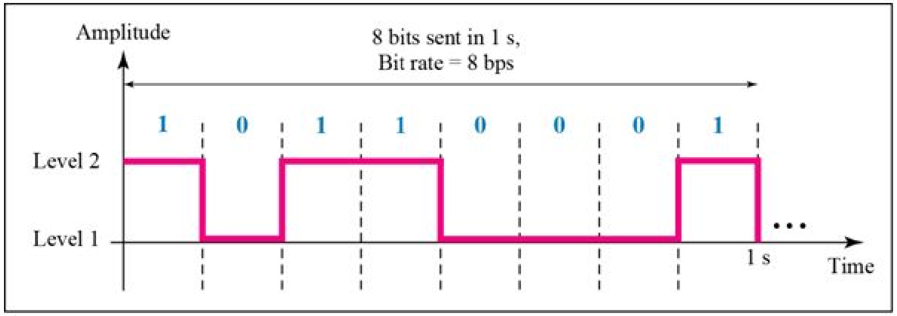
a. Bandwidth of a periodic signal



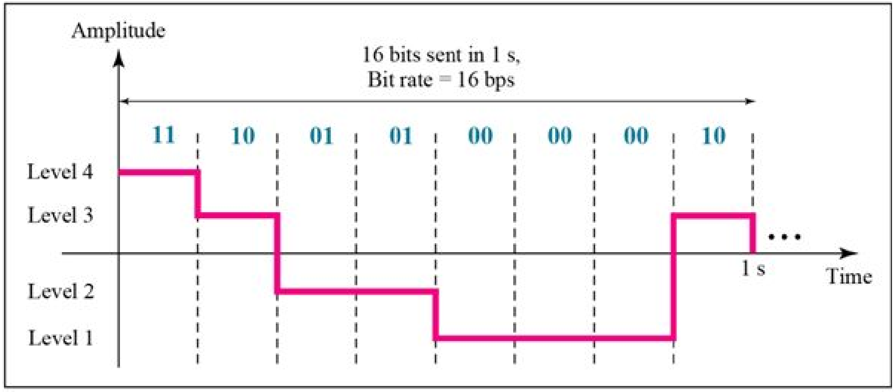
b. Bandwidth of a nonperiodic signal

Digital Signal

- In digital signal, 1 can be encoded as a positive voltage and a 0 as zero voltage
- May have more than two voltage levels when need to transfer more than one bit per level
- if a signal has L levels, each level can hold $\log_2 L$ bits.



a. A digital signal with two levels



b. A digital signal with four levels

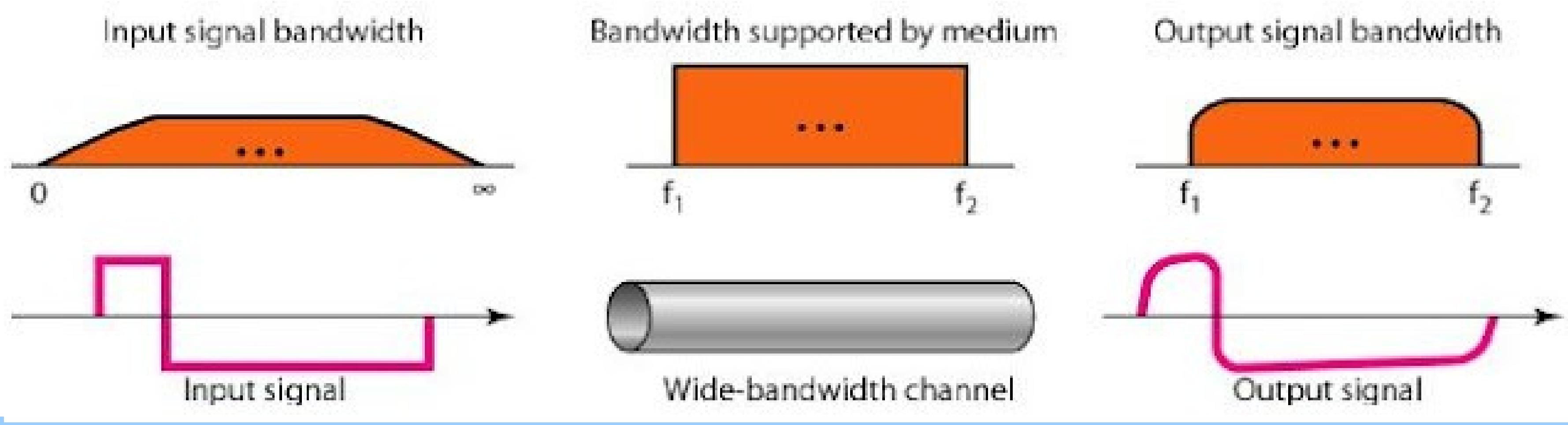
Bitrate

- Most digital signals are nonperiodic
- Bitrate is used to describe the signal (no. of bits sent per second)
- The bit length is the distance one bit occupies on the transmission medium.
  - Bit length =propagation speed x bit duration

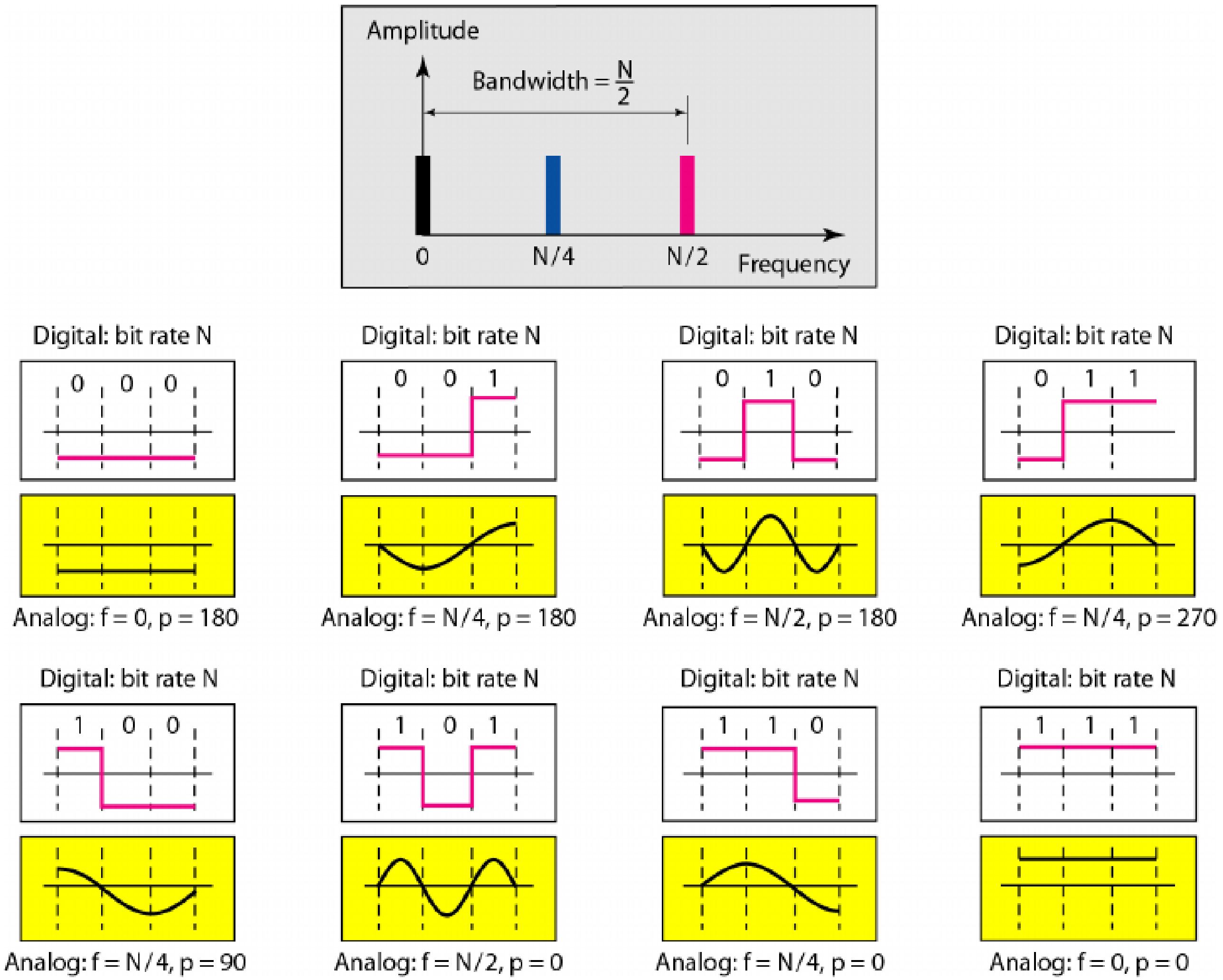
- Example
- A page is an average of 24 lines with 80 characters in each line. Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?
- A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency. We assume that each sample requires 8 bits. What is the required bit rate?

Transmission of Digital Signals

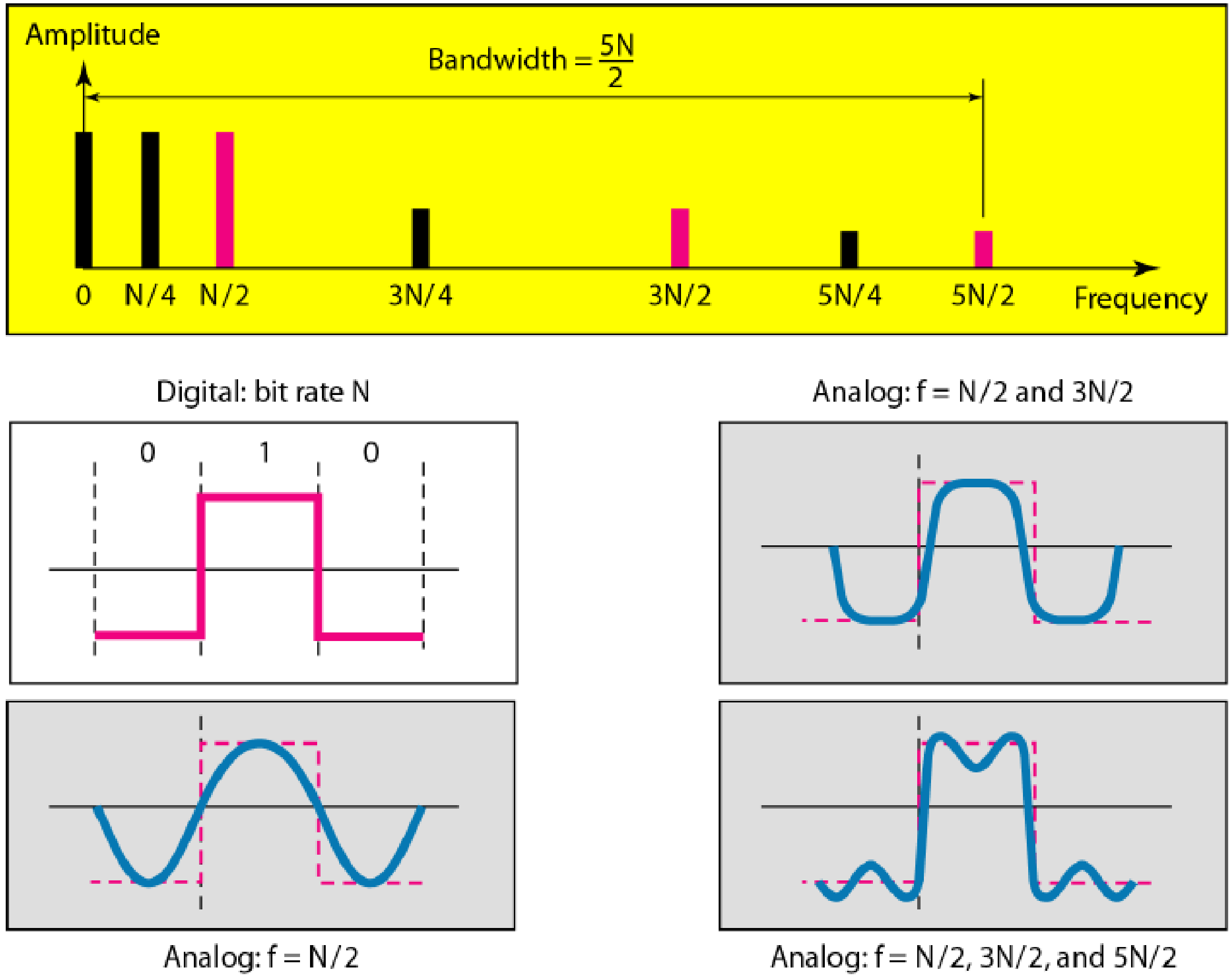
- **Baseband transmission** - sending a digital signal over a channel without changing the digital signal to an analog signal.
  - requires a low-pass channel (bandwidth starts from zero)
- Base-band transmission preserves the shape of the digital signal, only if we have a low-pass channel with an infinite or very wide bandwidth.



Rough Approximation

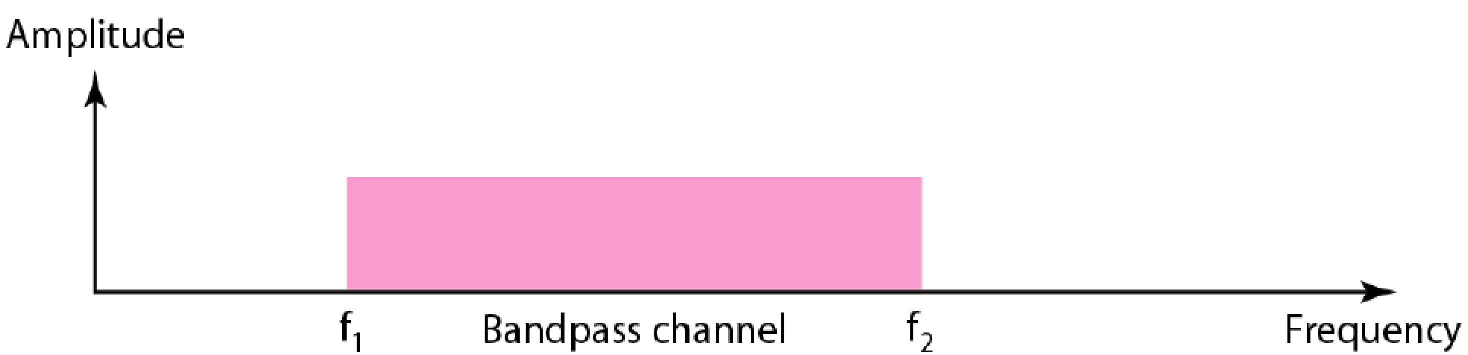


Better Approximation

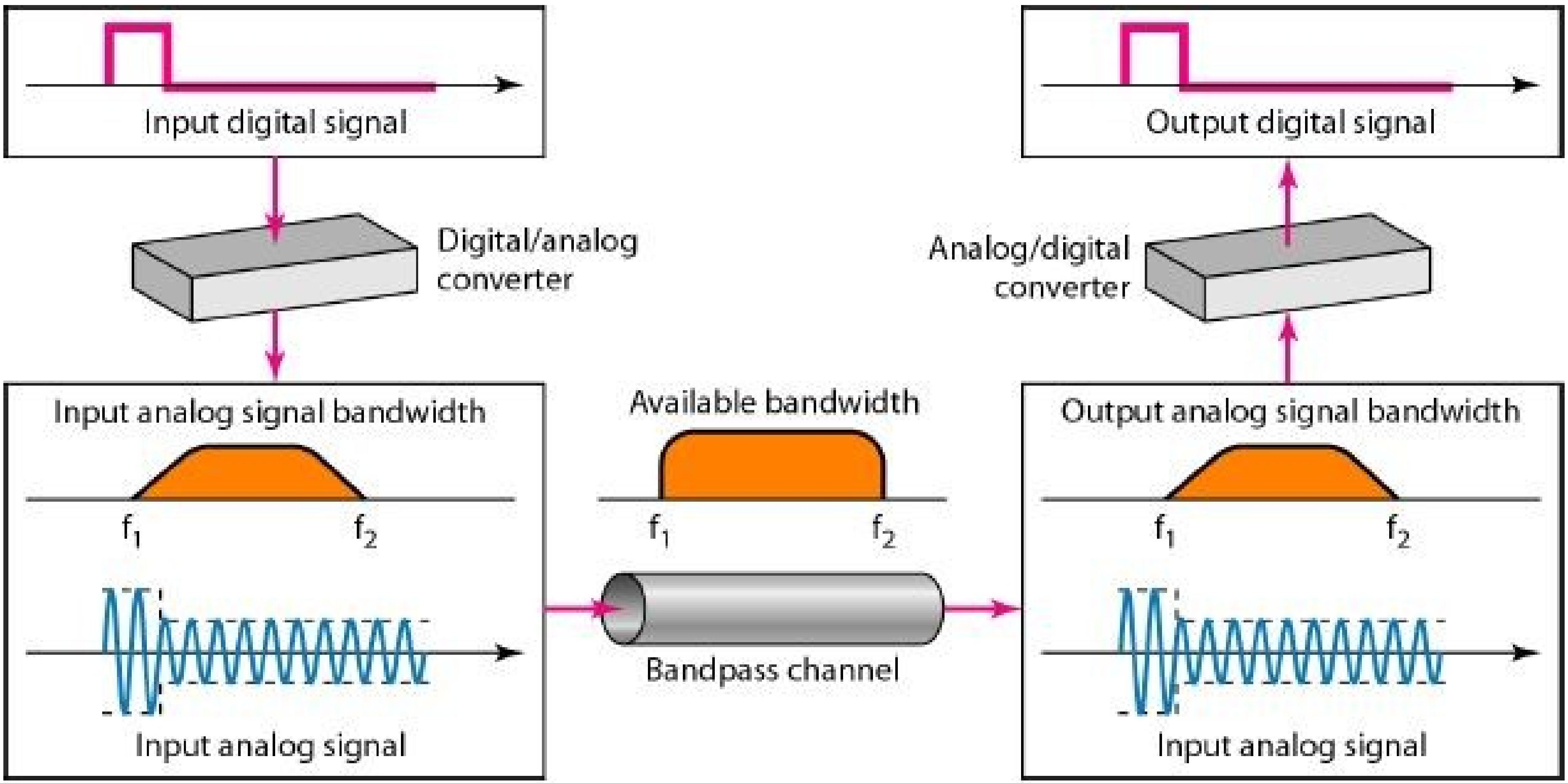


Transmission of Digital Signals

- **Broadband transmission** or modulation means changing the digital signal to an analog signal for transmission
- Modulation allows us to use a bandpass channel



Modulation of a digital signal



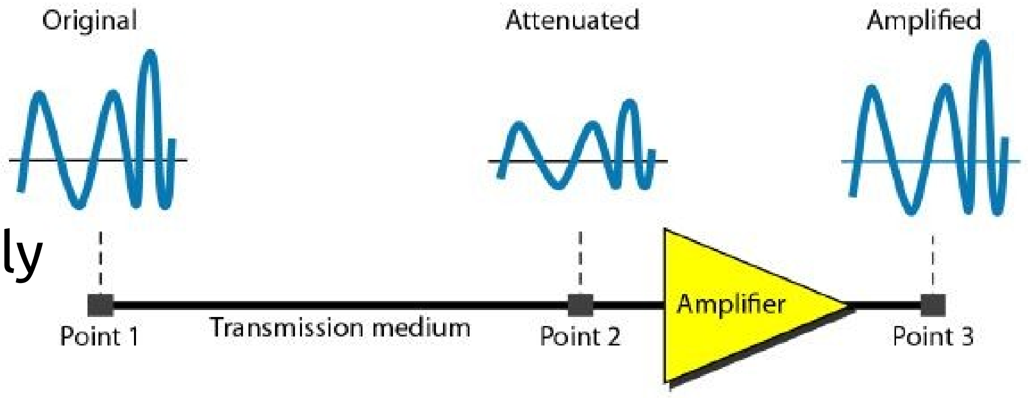
Transmission Impairment

- The signal at the beginning of the medium is not the same as the signal at the end of the medium
- Three causes of impairment
  - **Attenuation**
  - **Distortion**
  - **Noise**

Attenuation

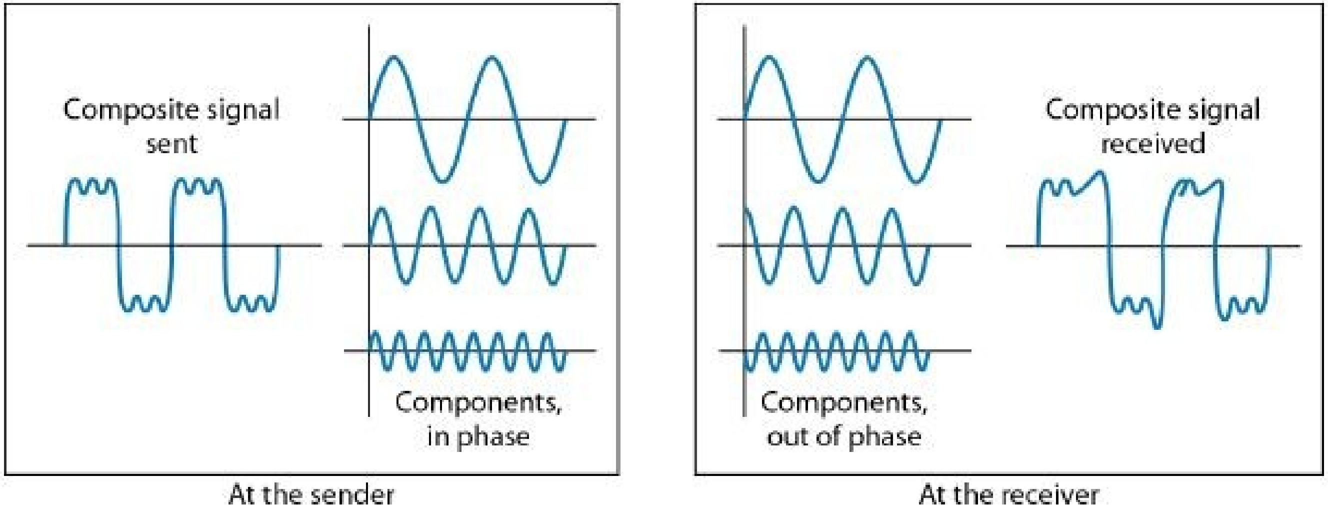
- Loss of energy in overcoming the resistance of the medium
- Amplifiers are used to amplify the signal
- The **decibel (dB)** measures the relative strengths of two signals or one signal at two different points
- The decibel is negative implies signal is attenuated
- The decibel positive implies signal is amplified.

- $dB = 10 \log_{10} P2/P1$
- P1, and P2 are the power of the signal at point 1 and 2 respectively



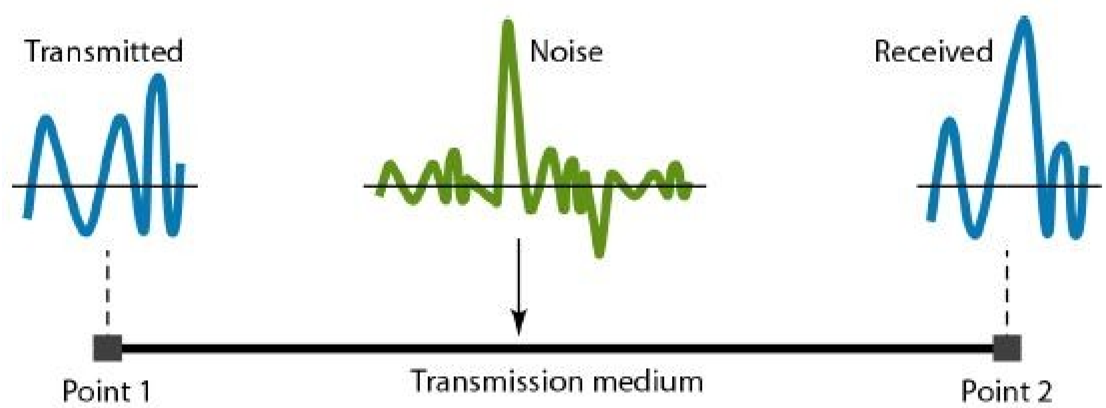
Distortion

- The signal changes its form or shape
- Can occur in a composite signal made of different frequencies
- Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration



Noise

- Corruprts the original signal
- Types : thermal noise, induced noise, crosstalk, and impulse noise
- Thermal noise is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter
- Induced noise comes from sources such as motors and appliances
- Crosstalk is the effect of one wire on the other
- Impulse noise is a spike that comes from power lines, lightning, and so on

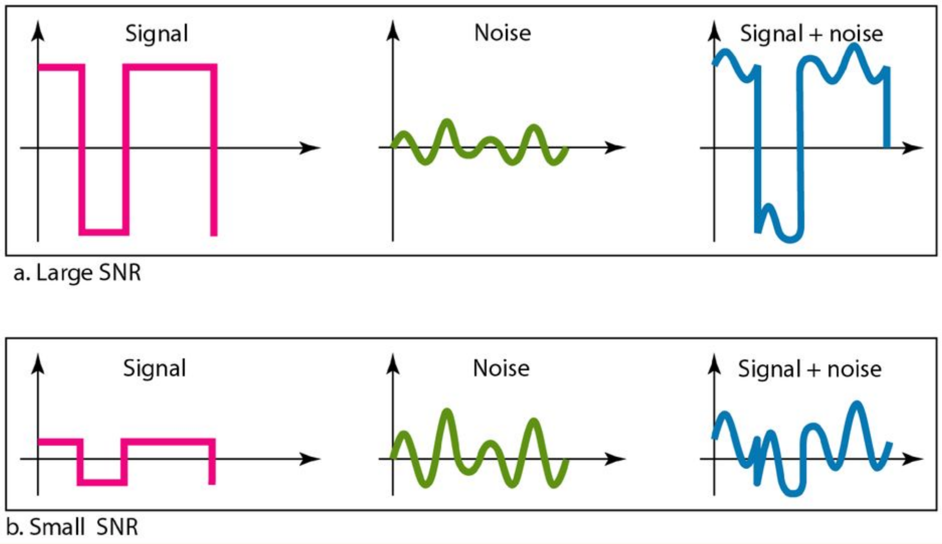


Signal-to-Noise Ratio (SNR)

- The signal-to-noise ratio is defined as
- SNR = average signal power/average noise power
- Because SNR is the ratio of two powers, it is often described in decibel units

$$\mathrm{SNR_{dB}} = 10 \log_{10}(\mathrm{SNR}).$$

Two cases of SNR: a high SNR and a low SNR



Data Rate Limit

- Data rate depends on three factors:
  - The bandwidth available
  - The level of the signals
  - The quality of the channel (the level of noise)
- Nyquist and Shannon have provided theoritical data limit on a noiseless channel and noisy channel respectively

Nyquist Bit Rate

- For a noiseless channel, the theoretical maximum bit rate is defined as
  - BitRate = 2 x bandwidth x $\log_2 L$

- Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. Then, The maximum bit rate can be calculated as
  - BitRate =2 x 3000 x $\log_2 2$ = 6000 bps
- We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?
  - 265,000 =2 X 20,000 X $\log_2 L$
  - $\log_2 L = 6.625$
  - $L=2^{6.625} = 98.7$ levels

Shannon Capacity

- Shannon capacity, to determine the theoretical highest data rate for a noisy channel is
  - Capacity = bandwidth X $\log_2$ (1 + SNR)
- No indication of the signal level means no matter how many levels we have, we cannot achieve a data rate higher than the capacity of the channel

Example

- We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as
- $C = B \log_2 (1 + SNR) = 3000 \log_2 (1 + 3162)$

$$= 3000 \log_2 3163$$

$$= 3000 \times 11.62 = 34{,}860 \text{ bps}$$

Example

- Assume that $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as
  - $SNR_{dB} = 10 \log_{10} SNR$
  - $\log_{10} SNR = 3.6$
  - $SNR = 10^{3.6} = 3981$
  - $C = B \log_2 (1 + 3981)$

Network Performance

- Bandwidth
- Throughput = (sender capacity * bandwith) /sec
- Lantency = propagation time + transmission time + queuing time + processing delay

Next : DigitalTransmission